

Antonio Pio Ricciardi

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My research focuses on the use of representation learning techniques in the reinforcement learning domain to enable zero-shot encoder-controller compositionality, combining the components of an agent, rather than retraining it from scratch. This allows us to create completely new agents capable of handling environment-task combinations never seen during training. Our work aims to pave the road toward a more accessible and flexible use of reinforcement learning.

Projects

Zero-Shot Stitching in Reinforcement Learning using Relative Representations | <https://arxiv.org/abs/2404.12917> (2024)

Using latent alignment techniques to enable zero-shot stitching between encoders and controllers of RL agents trained on environments with visual and task variations, to create new agents able to act on environment variations never seen during training.

Using entities dynamics from pixels | 2023

Using entities dynamics from pixels in visually complex Reinforcement Learning Environments as additional information for reinforcement learning agents, allowing faster training.

Shape modeling with reinforcement learning | 2022

Implement a reinforcement learning agent to model 2D and 3D shapes by moving vertices in space. Starting with an initial mesh, the agent's objective is to adjust the vertices to match a target mesh.

Experience

Teaching

Deep Learning & Applied AI, Machine Learning Courses @ Sapienza, University of Rome 2024

Reinforcement learning seminar: Fundamental concepts of RL, covering key topics such as policies, value functions, Q-learning and Deep Q-Learning. Additionally, developed and provided a comprehensive Jupyter Notebook containing exercises to reinforce the theoretical concepts discussed.

Introduction to Algorithms 2022

Teaching Assistant for bachelor of computer science course. Teaching assistant and member of the Exam Committee.

European Workshop On Reinforcement Learning

Poster Session (Non archival Paper) September 2023

Ricciardi Antonio Pio, Valentino Maiorca, Luca Moschella, and Emanuele Rodolà. "Zero-shot stitching in Reinforcement Learning using Relative Representations." In Sixteenth European Workshop on Reinforcement Learning. 2023. (<https://openreview.net/forum?id=4tcXslmfsS1>)

Phd Schools

Artificial Intelligence and Games - 5th International Summer School Cambridge (UK) June 2023

Summer school dedicated to the uses of artificial intelligence techniques in and for games. Participated in the game jam as a team of 6, won the 1st prize.

Artificial Intelligence and Games - 4th International Summer School Cambridge Chania (Crete) September 2022

Summer school dedicated to the uses of artificial intelligence techniques in and for games. Participated in the game jam as a team of 5.

Conference - The Ethics and Law of AI

Chamber of Deputies - Rome 21-22 November 2019

Chosen to be part of a delegation of 20 best students to represent La Sapienza at the Conference, held by Fondazione Leonardo to discuss about the technical, ethical and legal matters concerning Artificial Intelligence.

Education

Ph.D. in Computer Science 2021 - now

Sapienza, University of Rome

Research focus: Applications of representation learning techniques in the reinforcement learning domain to enable zero-shot encoder-controller compositionality.

MSc in Computer Science 2017-2020

Sapienza, University of Rome (Grade: 110 / 110 with honors, GPA: 3.92 / 4.00)

Thesis: Uncompressed inference for interpretable text classification.

Relevant Courses: Computer Graphics, Machine Learning, NLP, Advanced and Network Algorithms, Computer Vision.

Technical Skills

Research Interests: Reinforcement Learning, Deep Learning

Technologies: Python, PyTorch, OpenAI Gym, StableBaselines, Unity, Java, MATLAB, C#, Git, $\text{\LaTeX}$

Spoken Languages: Italian (Native), English (C1 - According to TOEFL iBT), French (beginner)

Achievements & Scholarships

- 1st place in the Game Jam at the 5th AI&Games Summer School, Cambridge (UK). (2023)
- Research **Grant** on *Interdisciplinary applications of Machine Learning*. (2023)
- Research **Grant** on *Structured prediction methods using spectral reinforcement learning*. (2021-2023)
- Lazio DiSCo **Scholarship**. (2018)